TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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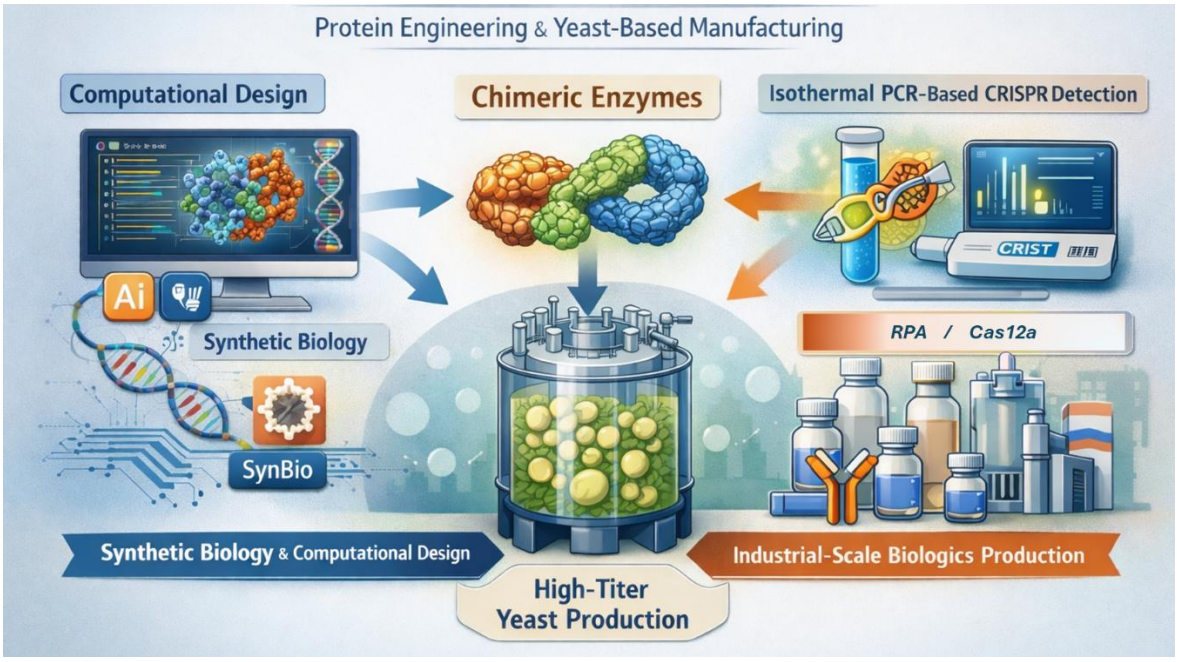
2. Technology Overview

Title of Proposed Technology	Development of Broad-Spectrum Lytic Enzymes for Universal Bacterial and Mycoplasma Lysis and Their Mass Production via Yeast Expression Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	Microbial and Enzyme Catalysis	Bio-Processes/Bio-Equipment	Bio-Engineering Technologies
Concept of Proposed Technology	Microbial contamination of biologics (protein and nucleic acid) production systems represents a major bioprocessing concern. Current approaches to decontaminate often use harsh chemicals that are not environmentally friendly, linger in bioreactors, and can harm/destroy the biologic product to be produced. For this reason, we propose to develop "**super antimicrobial enzymes**" capable of inducing complete bacterial cell lysis of a broad spectrum of microorganisms, including Mycoplasma, as well as both Gram-negative and Gram-positive bacteria. Production of these enzymes would be performed in yeast, thereby exploiting very high enzyme production capabilities. By achieving rapid and complete lysis, the technology enables the immediate release of high-purity genetic material for downstream analysis without complex pretreatment. Furthermore, these enzymes can be designed to serve as a robust contamination control solution in mammalian cell culture bioprocesses, ensuring safety and efficiency.			
Rationale for Joint R&D Project	The collaboration would leverage a Korean SME industry partner's advanced yeast expression system to achieve cost-effective, large-scale production of complex antimicrobial enzymes with optimal protein folding and activity. By integrating this manufacturing expertise with the development of novel lysis enzymes, the project addresses critical unmet needs in biopharmaceutical contamination management and rapid diagnostics. This vertical integration from design to mass production will accelerate the			

	commercialization of an antibiotic-free sterilization platform for the global market.
Key Words	1. Broad-Spectrum Lytic Enzymes. 2. Mycoplasma Elimination. 3. Yeast-Based Mass Production. 4. Detection of DNA Biomarkers Post-cell Lysis. 5. Bioprocess Contamination Control.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	The R&D objectives for this proposal center on a multi-disciplinary framework that integrates advanced protein engineering with a high-efficiency yeast-based manufacturing platform. By leveraging computational design and synthetic biology, the research aims to produce chimeric enzymes with high-titer yields in yeast systems, specifically optimized for the rapid detection of microbial contamination and industrial-scale biologics production (see **Schematic illustration**).  **Schematic illustration.** R&D objectives integrating protein engineering with yeast-based manufacturing and CRISPR-based contamination detection. The left panel illustrates computational design and synthetic biology tools used to design and optimize chimeric enzymes. These engineered enzymes are then produced at high titer using a yeast-based manufacturing platform (central bioreactor). The right panel depicts an isothermal PCR-based CRISPR detection system, representing rapid and sensitive microbial contamination monitoring. Finally, the bottom panel highlights the ultimate goal of industrial-scale biologics production, demonstrating how the engineered enzymes and detection platform support scalable biomanufacturing.			

1. Enzyme Engineering & "Super Antimicrobial Enzyme Candidates" Design

The primary research goal involves the rational design of modular, chimeric enzymes that combine specific cell-binding domains with potent catalytic domains.[1] These engineered "super enzymes" must be structurally optimized to penetrate the complex, multi-layered cell envelopes of Gram-negative bacteria,[2] the thick peptidoglycan of Gram-positive species,[3] and the flexible lipid membranes of Mycoplasma.[4] This design phase requires rigorous computational modeling, including Artificial intelligence (AI)[5] and directed evolution to ensure the enzymes maintain high lytic activity within the specific pH and temperature ranges of a mammalian bioreactor without interfering with the therapeutic biologics being produced.

2. Yeast Expression System Optimization

A secondary objective focuses on the development of a robust yeast-based production platform, likely using *Pichia pastoris* or *Saccharomyces cerevisiae*.[6-7] The research team will perform strain engineering to maximize the secretion of these antimicrobial proteins into the culture media, exploiting yeast's natural ability to produce complex proteins at high titers. This phase includes the optimization of fermentation parameters and the creation of a streamlined purification process to ensure the enzymes can be manufactured cost-effectively and at a scale suitable for industrial bioprocessing applications.

3. Lysis Efficiency & Analytical Validation

The third objective centers on validating the technology's dual-use potential, integrating robust contamination management with streamlined rapid diagnostics. The R&D framework employs kinetic profiling to characterize the rate of bacterial lysis and evaluate the integrity of the resulting genetic material. A primary goal is to demonstrate that this enzymatic platform enables the immediate liberation of high-quality DNA or RNA biomarkers, facilitating seamless integration with downstream isothermal amplification-based CRISPR detection.[8-9] By eliminating the need for harsh chemical pretreatments and exhaustive purification steps, this approach preserves the sensitivity of CRISPR-based diagnostic tools, such as the targeting of specific nucleic acid sequences by Cas12a or Cas13 effectors, while ensuring the entire process remains rapid and biologically compatible.

4. Safety and Environmental Impact Assessment

The research aims to establish the safety and "clearability" of these enzymes within a mammalian cell culture environment, utilizing compatibility assays to confirm they are non-toxic to CHO or HEK293 cell lines while verifying their seamless removal during standard downstream purification.[10] By implementing these antimicrobial enzymes, early-stage microbial contamination in biologics manufacturing can be rapidly suppressed or eliminated before it compromises the culture, significantly mitigating the risks of batch failure, unplanned process shutdowns, and costly production delays.[11] This intervention enhances overall process robustness and manufacturing reliability, resulting in more consistent product quality and reduced operational risk.[12] Furthermore, by demonstrating the biodegradable nature of these enzymes in contrast to persistent chemical decontaminants, the R&D efforts will validate the environmental and operational advantages of transitioning to a biologically based contamination management system.

R&D Target Specifications	**1. Enzyme Performance & Lysis Kinetics** The primary goal is to achieve rapid microbial inactivation that outperforms traditional chemical methods while remaining gentle on the biologic product. • Lytic Range: Demonstrated efficacy against a broad spectrum, including Gram-positive (*B. subtilis*), Gram-negative (*E. coli, P. aeruginosa*), and cell-wall-deficient *Mycoplasma* species. • Lysis Speed: Achievement of > 99% (2-log reduction) microbial cell death within 30 minutes of enzyme introduction. • Operating Window: Maintenance of > 80% maximal lytic activity within standard bioreactor parameters (pH 6.8–7.5 and temperature 31°C–37°C). **2. Yeast Production & Expression Yields** The yeast platform must provide a cost-effective alternative to synthetic antimicrobial peptides or purified animal-derived enzymes. • Target Titer: Achievement of at least 2–5 g/L of secreted chimeric enzyme in fed-batch fermentation using *Pichia pastoris.* • Purity & Recovery: Final enzyme purity of > 90% following a simplified two-step purification process (e.g., tangential flow filtration (TFF) and His-tag Affinity Chromatography), with a total recovery yield of > 70%. • Stability: The purified enzyme must remain stable and active for at least 6 months when stored at 4°C or -20°C in a stabilized formulation. **3. Diagnostic Sensitivity & Bioprocess Integration** The "instant-release" workflow must facilitate high-sensitivity detection without the need for manual sample preparation. • Detection Sensitivity: Successful detection of contamination at levels as low as 10–1000 CFU/mL using the enzyme-lysed sample directly in RPA-CRISPR assays. • Nucleic Acid Integrity: Post-lysis DNA/RNA must show a high DNA/RNA Integrity Number, confirming that the lytic enzymes do not possess non-specific nuclease activity. • Product Safety: Zero detectable degradation of the therapeutic biologic (monoclonal antibody or viral vector) as measured by SEC-HPLC and SDS-PAGE following a 24-hour co-incubation with the enzymes. **4. Environmental & Operational Metrics** The technology must simplify the facility's environmental footprint and reduce the risk of batch loss. • Biodegradability: Confirm that the active structure is neutralized and degraded into non-toxic peptides and amino acids within 48 hours in standard wastewater treatment conditions, eliminating the need for chemical neutralization. • Risk Mitigation: A projected > 50% reduction in batch failure rates due to microbial contamination through early-stage enzymatic suppression.
R&D Plan	**Task 1: "Super Antimicrobial Enzyme Candidates" Design (Months 0-6, TRL3)** The initial task focuses on the architectural creation of chimeric lysins.[13] Research begins with the *in silico* screening of diverse catalytic domains, including endolysins for peptidoglycan hydrolysis,

phospholipases for membrane disruption, and specialized antimicrobial peptide sequences to target bacteria. These disparate modules are strategically fused using flexible linkers to ensure independent folding and synergistic activity. A critical component of this phase involves the optimization of Cell Wall Binding Domains (CBDs) to enhance the affinity of the enzyme for the complex Gram-negative outer membrane. Once the optimal primary structures are determined, the genetic sequences will undergo comprehensive codon optimization specifically tailored for the translational machinery of the selected Korean yeast host. This ensures that the high-complexity "super antimicrobial enzymes" are compatible with the host's secretion pathways, minimizing intracellular aggregation and maximizing the yield of functional proteins.

- **Modeling**: We will use AI-driven protein structure prediction (e.g., AlphaFold3)[14] to model the docking efficiency of various CBDs against specific bacterial cell wall ligands.
- **Modular Assembly**: We will construct a library of fusion proteins combining enzymatically active domains (EADs, such as endopeptidases and muramidases) with CBDs tailored for the unique peptidoglycan cross-links of Gram-positives and the outer membrane lipopolysaccharides of Gram-negatives.

Deliverable: A full proposal prepared jointly by RPI and a Korean SME.

Milestone: AI-Driven design and structural validation of chimeric "super antimicrobial enzyme candidates".

Task 2: Yeast Expression System Optimization (Months 7–18, TRL 4–5)

Following the design phase, the project transitions into the development of a robust manufacturing platform using high-performance yeast expression systems. The synthetic gene constructs will be integrated into the yeast genome through precise homologous recombination, followed by the selection of "super-producers" based on copy number and secretion efficiency. To ensure industrial viability, we will develop a high-yield production strain using *Pichia pastoris*.

- **Strain Engineering**: We will employ CRISPR-based genomic integration or alternative high-efficiency systems to incorporate multiple copies of the enzyme-encoding genes into the yeast genome, using either inducible (e.g., AOX1) or constitutive (e.g., GAP) promoters for precise expression control.[15] To maximize extracellular yields, we will perform systematic optimization of secretion signals, such as the alpha-mating factor leader sequence, to enhance protein trafficking and secretion efficiency within the yeast host.[16]
- **Fermentation Scale-up**: R&D will transition from shake-flask cultures to 10-L fed-batch bioreactors, optimizing dissolved oxygen, carbon sourcing (methanol vs. glycerol), and nitrogen feed to achieve ultra-high titers.
- **Streamlined Purification**: We will develop a simplified downstream process utilizing His-tag affinity chromatography or tangential flow filtration (TFF), aiming for >90% purity to prevent yeast-host cell protein (HCP) carryover into the final application.

Deliverable: Optimized chimeric sequences; Synthetic plasmid library; Optimized yeast expression system.

Milestone: Optimization of fermentation parameters in up to a 10-L batch or fed-batch bioreactor.

Task 3: Lysis Efficiency & Analytical Validation (Months 7-28, TRL 4–5)

This task bridges the gap between decontamination and diagnostic utility by characterizing the enzymes' ability to release "ready-to-analyze" genetic material. This phase is dedicated to validating the practical efficacy of the engineered enzymes within real-world diagnostic and biomanufacturing scenarios. The research team will standardize optimized lysis protocols to facilitate an "instant-release" workflow, ensuring that target bacterial DNA/RNA biomarkers are immediately accessible for isothermal amplification, such as Recombinase Polymerase Amplification (RPA), and subsequent CRISPR-Cas12/13a detection. By eliminating the need for traditional, time-consuming extraction kits, this approach preserves the sensitivity of the genetic biomarkers, allowing for a seamless, sample-to-signal diagnostic platform that is both rapid and biologically compatible.

- **Kinetic Profiling**: We will perform time-kill assays to determine the minimum inhibitory concentration (MIC) and the time required for complete (> 99%) lysis across target species.
- **Diagnostic Integration**: We will validate the compatibility of the lysate with isothermal amplification (e.g. RPA). The objective is a "sample-to-signal" workflow where enzymatic lysis provides the DNA/RNA template for CRISPR-Cas12/13a detection without intermediate purification.
- **Biomarker Integrity**: This preservation of nucleic acid integrity is critical for ensuring the robust performance of downstream isothermal amplification-based CRISPR detection, as it maintains the accessibility of target sequences for Cas12/13 effectors.

Deliverable: Provisional patent submitted; joint publication submitted.

Milestone: Characterization of lysis kinetics & MIC thresholds; Validation of "Instant-Release" nucleic acid integrity; Pilot "Sample-to-Signal" diagnostic integration

Task 4: Safety and Environmental Impact Assessment (Months 29–36, TRL 5)

The final task validates the integration of the enzymes into existing biologics workflows and confirms their "green" credentials. The technology will be rigorously tested within the context of mammalian cell culture, specifically targeting the selective elimination of contaminants in CHO and HEK293 cell lines. These studies focus on establishing a "safety window" where the enzyme effectively eradicates all microbial threats while remaining inert toward the mammalian host cells and the therapeutic proteins they produce. Comprehensive stability assays will be performed to ensure the enzyme maintains its potency under various storage conditions and throughout the duration of a typical bioreactor run, ultimately proving its value as a dual-purpose tool for both rapid detection and active contamination control.

- **Specificity Screening**: Through high-throughput assays and Surface Plasmon Resonance (SPR) binding analysis, we will verify that these enzymes selectively bind to target bacteria without interacting with the mammalian host cells used for biologics production.
- **Mammalian Cell Compatibility**: We will conduct MTT and LDH cytotoxicity assays on CHO and HEK293 cell lines to ensure the enzymes do not impede growth or metabolic health during co-incubation.

	• **Clearance Validation**: Using enzyme-linked immunosorbent assays (ELISA) and mass spectrometry, we will demonstrate that the antimicrobial enzymes are effectively removed during standard Protein A or Ion-Exchange chromatography steps in the biologics purification train. • **Biodegradability Testing**: We will measure the half-life of the enzymes in aqueous environments and bioreactor waste streams, documenting their degradation into non-toxic amino acids to support their superiority over persistent chemical decontaminants. **Deliverable:** Korean and U.S. Patent applications filed. **Milestone**: Validation of mammalian cell compatibility & specificity; Demonstration of downstream clearance; Verification of biodegradability & environmental safety.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI serves as the Lead Innovation Hub, driving the upstream "Design-Build-Test" cycle. • **Enzyme Design & AI Modeling**: Designing the “super antimicrobial enzymes” utilizing advanced computational tools and Artificial Intelligence (e.g., AlphaFold3). RPI, together with the Korean SME, is responsible for designing the molecular architecture for “super antimicrobial enzymes” using its CBD and EAD expertise to ensure broad-spectrum activity against Gram+, Gram-, and Mycoplasma. (**Task 1**). • **Biophysical & Kinetic Validation**: Using Surface Plasmon Resonance (SPR) to rigorously characterize binding kinetics and verify high-affinity specificity toward target microbes, while confirming the absence of "off-target" interactions with therapeutic biologics or mammalian host cell surfaces (**Task 3**). • **Diagnostic Kit Integration (CRISPR/RPA)**: Developing the finalized diagnostic workflow. The RPI is responsible for integrating the "instant-release" lysate into a user-friendly, isothermal amplification-based CRISPR detection kit (Cas12/13a) (**Task 3**). • **Mammalian Cell Studies**: The integration of the super antimicrobial enzymes into mammalian cell culture processes will be iteratively assessed along with the Korean SME partner (**Task 4**).
Expected R&R of Korean SME Partner	The Korean SME acts as the Translational & Commercialization Partner, focusing on the "Apply-Scale-Commercialize" cycle. • **Enzyme Design & AI Modeling**: Designing the “super antimicrobial enzymes.” Korean SME, in collaboration with RPI, will contribute its proprietary antimicrobial technologies, including but not limited to antimicrobial peptides, to support the “super antimicrobial enzymes” design (**Task 1**). • **Fundamental Yeast Engineering**: Establishing the initial high-expression strains using CRISPR-Cas9 genomic integration and optimizing secretion signal sequences (e.g., alpha-mating factor) for maximum titer (**Task 2**). • **Pilot Production**: Transitioning the yeast production protocols from laboratory-scale (shake flask) to industrial-scale 1L–10L batch or fed-batch fermentation, focusing on cost-effective

	manufacturing for the global market (**Tasks 2 & 4**). • **Industrial Compatibility & Field Testing**: Conducting real-world validation within Korean biologics manufacturing facilities to ensure the enzymes effectively mitigate batch failure risks in industrial bioreactors (**Task 4**).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The primary outcome of this research will be the realization of a first-in-class, broad-spectrum enzymatic lysis platform that transcends the limitations of current chemical and mechanical cell-disruption methods. By providing a "single-step" lysis solution for Gram-negative bacteria, Gram-positive bacteria, and Mycoplasma, this technology will fundamentally accelerate the speed of microbial identification in mammalian cell culture bioprocesses. In the diagnostic field, this translates to drastically reduced "sample-to-answer" times, allowing for near-instantaneous detection of contaminants in critical environments. Beyond rapid testing, the project will establish a new gold standard for "clean-label" bioprocessing. By replacing or supplementing traditional antibiotics with highly selective, biodegradable enzymes, the industry can mitigate the rising threat of antimicrobial resistance (AMR) while ensuring that animal cell cultures remain pristine and productive.[17] This move toward antibiotic-free manufacturing is expected to enhance the regulatory compliance and safety profiles of high-value biologics, such as monoclonal antibodies and gene therapies. Furthermore, the strategic partnership with a Korean company in yeast expression technology will generate a powerful economic and technical synergy. The project will prove the viability of using advanced yeast strains for the low-cost, high-volume production of complex, multi-domain proteins, which has historically been a bottleneck in the commercialization of therapeutic and industrial enzymes.[18] This scalable manufacturing model will not only secure a competitive edge in the global enzyme market but also provide a robust blueprint for future joint R&D ventures between molecular engineering labs and industrial manufacturing hubs. Ultimately, the impact of this project extends beyond mere contamination control; it offers a versatile biotechnological toolset that empowers researchers and manufacturers to maintain the highest levels of biological integrity, thereby fostering innovation across the entire life sciences ecosystem. **Team** **Jonathan S. Dordick, Ph.D.,** is Institute Professor of Chemical and Biological Engineering at RPI, with a joint appointment in the Department of Biological Sciences. He also serves as Vice President for Strategic Alliances and Translation. He received his B.A. degree in Biochemistry and Chemistry from Brandeis University and his Ph.D. in Biochemical Engineering from the Massachusetts Institute of Technology. At RPI, he has served as the Vice President for Research, founding Co-Director of the Rensselaer-Mount Sinai Center for Engineering and Precision Medicine, Director of the Center for Biotechnology & Interdisciplinary Studies, and Department Chair. Dr. Dordick has made foundational

	contributions to protein science and enzyme technology, biomanufacturing of therapeutic proteins and nucleic acids, and drug discovery. Dr. Dordick has received numerous awards and honors, including election to the National Academy of Engineering and National Academy of Inventors, and receipt of numerous society awards. He has founded several companies, and published over 460 papers and is an inventor/co-inventor on over 50 patents. **Seok-Joon Kwon, Ph.D.,** is a Senior Research Scientist at RPI with two decades of expertise bridging molecular biology, protein engineering, glycobiology, and nanobiotechnology. His research focuses on developing high-throughput screening assays, CRISPR-based systems, and combinatorial methods for drug discovery. He has a portfolio of over 60 publications, including papers in prestigious peer-reviewed journals, book chapters, and patents.

6. References

(1) Wu, X., Kwon, S. J., Kim, J., Kane, R. S. & Dordick, J. S. Biocatalytic Nanocomposites for Combating Bacterial Pathogens. *Annu Rev Chem Biomol Eng* **8**, 87-113 (2017).

(2) Briers, Y. et al. Engineered endolysin-based "Artilysins" to combat multidrug-resistant gram-negative pathogens. *mBio* **5**, e01379-01314 (2014).

(3) Yang, H., Zhang, H., Wang, J., Yu, J. & Wei, H. A novel chimeric lysin with robust antibacterial activity against planktonic and biofilm methicillin-resistant Staphylococcus aureus. *Sci Rep* **7**, 40182 (2017).

(4) Hayashi, K. et al. The effects of magainin 2-derived and rationally designed antimicrobial peptides on Mycoplasma pneumoniae. *PLoS One* **17**, e0261893 (2022).

(5) Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583-589 (2021).

(6) Ahmad, M., Hirz, M., Pichler, H. & Schwab, H. Protein expression in Pichia pastoris: recent achievements and perspectives for heterologous protein production. *Appl Microbiol Biotechnol* **98**, 5301-5317 (2014).

(7) Kim, M. J., Sung, B. H., Kim, H. J., Sohn, J. H. & Bae, J. H. Production of autolysis-proof Kex2 protease from Candida albicans in Saccharomyces cerevisiae for in vitro processing of fusion proteins. *Appl Microbiol Biotechnol* **106**, 7063-7072 (2022).

(8) Gootenberg, J. S. et al. Nucleic acid detection with CRISPR-Cas13a/C2c2. *Science* **356**, 438-442 (2017).

(9) Chen, J. S. et al. CRISPR-Cas12a target binding unleashes indiscriminate single-stranded DNase activity. *Science* **360**, 436-439 (2018).

(10) Gronemeyer, P., Ditz, R. & Strube, J. Trends in Upstream and Downstream Process Development for Antibody Manufacturing. *Bioengineering* **1**, 188-212 (2014).

(11) Fratz-Berilla, E. J. et al. Impacts of intentional mycoplasma contamination on CHO cell bioreactor cultures. *Biotechnol Bioeng* **116**, 3242-3252 (2019).

(12) Gravante, F. et al. Preventing Microorganism Contamination in Starting Active Materials for Synthesis from Global Regulatory Agencies: Overview for Public Health Implications. *Microorganisms* **13**, 1595 (2025).

(13) Kim, D. et al. Modular Assembly of Unique Chimeric Lytic Enzymes on a Protein Scaffold Possessing Anti-Staphylococcal Activity. *Biomacromolecules* **20**, 4035-4043 (2019).

(14) Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493-500 (2024).

(15) Liu, Q. et al. CRISPR-Cas9-mediated genomic multiloci integration in Pichia pastoris. *Microb Cell Fact* **18**, 144 (2019).

(16) Chahal, S. et al. Structural characterization of the alpha-mating factor prepro-peptide for secretion of recombinant proteins in Pichia pastoris. *Gene* **598**, 50-62 (2017).

(17) Brechun, K. E., Forschle M., Schmidt M., Kranz H., Inducible complementation for antibiotic-free plasmid-based biomanufacturing in industrially relevant strains, *Front. Ind. Microbiol.* **3**, 1725733 (2026).

(18) Madhavan, A. et al. Customized yeast cell factories for biopharmaceuticals: from cell engineering to process scale up. *Microb Cell Fact* **20**, 124 (2021).

Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization